BITS BSc Computer Science Study Project Phase 1 Submission

Group - 116

Project Mentor: Prof. Raj Kumar K

Project Name: SmartCloud Secrets Manager with AI-Based Risk Analysis and Auto-Rotation

Group Members:

Nymish Kashivishwanath	2023EBCS686
Shrinibas Mahanta	2023EBCS695
Prem Bhavesh Shah	2023EBCS615
Abhinav Nagar	2023EBCS623

Study Project – Phase 1 Document

Project Title: SmartCloud Secrets Manager
Team Members:

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Project Advisor / Supervisor: Prof. Raj Kumar K
Date of Submission:

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1. Project Idea Summary

Title of Project

SmartCloud Secrets Manager with AI-Based Risk Analysis and Auto-Rotation

Abstract

In modern cloud environments, managing sensitive credentials such as API keys, database passwords, and access tokens is critical for security and system reliability. Traditional secret management systems, including existing cloud-native solutions, mainly focus on secure storage and retrieval but lack proactive intelligence and cross-cloud flexibility.

This project proposes the development of a SmartCloud Secrets Manager that integrates artificial intelligence-based risk analysis, automated secret rotation, and multi-cloud interoperability. The system will securely store secrets using encryption, monitor access patterns, detect risky behaviors, and automatically rotate compromised or weak credentials. It will also support managing secrets across multiple cloud platforms such as AWS, Azure, and GCP through a unified interface.

The proposed solution aims to reduce security breaches, minimize manual administrative effort, and improve overall cloud security management. This system will be beneficial for developers,

administrators, and organizations seeking a more intelligent, scalable, and secure approach to secrets management.

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2. Project Background and Motivation

Problem Statement

Modern cloud applications rely heavily on secrets such as passwords, tokens, and encryption keys. Improper management of these secrets can lead to data breaches, unauthorized access, and system failures. Existing solutions primarily provide storage and access control but do not actively analyze risks or automate recovery.

Developers and system administrators are affected due to manual secret handling, lack of centralized multi-cloud control, and delayed detection of security threats.

Current solutions are insufficient because:

- They lack intelligent risk assessment.
- They require manual rotation.
- They are mostly limited to single-cloud platforms.

Motivation

This problem is worth solving because security breaches caused by leaked credentials are increasing in cloud environments. Improving secret management can significantly reduce organizational risk.

From an academic perspective, this project applies cloud security concepts, distributed systems, and machine learning. Technically, it enhances automation and intelligent monitoring. Personally and academically, this project aligns with course objectives and builds industry-relevant skills.

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3. Educational Value and Course Alignment

Relevance to Course Objectives

This project aligns with:

- Cloud infrastructure management
- Security in cloud computing

- Distributed systems
- DevOps practices
- Automation and monitoring

It provides practical implementation of theoretical concepts taught in the course.

Learning Outcomes

After completing this project, students will gain:

- Experience in cloud service integration
- Knowledge of encryption and security practices
- API and backend development skills
- Exposure to machine learning basics
- System design and documentation skills
- DevOps and deployment experience

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4. Objectives

Primary Objectives

- To design and develop a secure secrets management system
- To implement AI-based risk analysis
- To support automated secret rotation
- To enable multi-cloud secret management
- To provide a centralized management dashboard

Secondary Objectives

- To implement logging and auditing mechanisms
- To improve system usability
- To support CLI-based access
- To generate security reports

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5. Research and Analysis / Literature Survey

Existing Solutions

Existing tools include:

- AWS Secrets Manager
- HashiCorp Vault
- Azure Key Vault
- Google Secret Manager

Limitations:

- Limited AI integration
- Mostly manual rotation
- Vendor lock-in
- Limited predictive security features

Functional Requirements

The system should:

- Store secrets securely using encryption
- Authenticate users
- Retrieve and update secrets
- Analyze access patterns
- Generate risk scores
- Rotate secrets automatically
- Support multiple cloud providers
- Provide dashboards and reports
- Log all activities

Non-Functional Requirements

- **Usability:** User-friendly interface
- **Performance:** Fast secret retrieval
- **Security:** Strong encryption and access control
- **Scalability:** Support growing users and data
- **Maintainability:** Modular architecture

Feasibility Analysis

- **Technical Feasibility:** High (existing cloud SDKs and libraries available)
- **Time Feasibility:** Achievable within project timeline
- **Resource Constraints:** Uses open-source tools and free-tier cloud services

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6. Project Scope and Expected Deliverables

Scope Definition

Included:

- Secure storage
- AI risk analyzer
- Auto-rotation
- Multi-cloud integration
- Web dashboard

Excluded:

- Enterprise-scale deployment
- Commercial-grade compliance certifications

Assumptions and Constraints:

- Internet connectivity
- Limited cloud budget
- Academic usage

Deliverables (Phase 1)

- Project proposal document
- Problem definition
- Objectives and scope
- Literature review
- High-level system design

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7. Preliminary Project Timeline and Milestones

Phase	Activity	Duration
Phase 1	Research and Planning	Weeks 1–3
Phase 2	System Design	Weeks 4–5
Phase 3	Backend Development	Weeks 6–9

Phase 4	AI Module	Weeks 10–11
Phase 5	Frontend Development	Weeks 12–13
Phase 6	Testing & Deployment	Weeks 14–15
Phase 7	Documentation	Week 16

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8. Team Structure and Collaboration

Roles and Responsibilities

Name	Role	Responsibilities
Nymish	Team Lead & DevOps Engineer	Project coordination, CI/CD setup, deployment, monitoring, cloud configuration
Shrinibas	Backend & Infrastructure Engineer	API development, database management, encryption, cloud integration
Prem	Frontend Engineer	UI/UX design, web dashboard development, user interface implementation
Abhinav	AI & Security Engineer	AI risk analyzer development, security testing, risk scoring, alert system

Collaboration Tools

- GitHub (Version Control)
- Google Docs
- Trello / Notion
- Slack / WhatsApp

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9. Risk and Challenge Analysis

Identified Risks

Risk	Description
Technical Complexity	Multi-cloud integration difficulty
Security Risks	Misconfiguration vulnerabilities
Time Constraints	Limited semester time
API Limitations	Cloud service restrictions

Mitigation Strategies

Risk	Mitigation
Technical Complexity	Modular design
Security Risks	Code reviews & testing
Time Constraints	Proper scheduling
API Limitations	Use mock services

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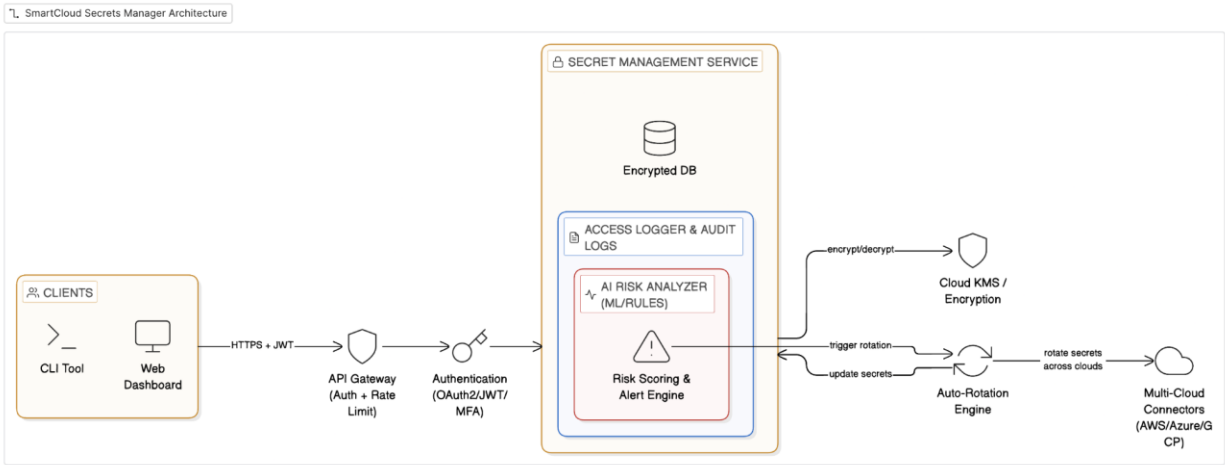
10. High Level Diagram



11. Supervisor Review and Approval

Advisor Feedback:

Supervisor Comments: The team has listed all major deliverables for Phase 1 in this document. Document approved.

Recommendations: Recommended to submit on time as per BITS timeline.

Signature:

Date: 2/2/2026